Cooperative Policy Gradient with Lossy Communication in General Stochastic Games

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Abstract

We extend the evidence-weighted policy gradient (EW-PG) and opponent-shaping PG (LOLA-PG) frameworks to cooperative settings where agents form coalitions and communicate policies through lossy channels. Three key contributions: (1) We formalize the *self-knowledge problem*: an agent's ability to communicate its policy is bounded by its evidence quality, connecting the $\mathcal{O} \to \Pi$ inference gap to information-theoretic communication limits. (2) We show that communication introduces a bias decomposable into self-knowledge loss and channel loss, with convergence preserved under annealed cooperation. (3) We prove that coalition formation enlarges the basin of attraction through a cooperative reinforcement mechanism dual to LOLA's spectral reinforcement. The evidence weight w_i serves triple duty: gradient quality, self-knowledge, and communication quality — all manifestations of the same underlying information content. This completes the formalization of all five components of the Ω-gradient.

1 The Self-Knowledge Problem

1.1 The $\mathcal{O} \to \Pi$ Gap

In the standard PG framework, agent i has policy $\pi_i \in \Pi$ and computes gradient estimates $\hat{v}_i$ to improve it. The implicit assumption is that the agent *knows* its own policy. But in the Ω-framework, agents operate in observation space $\mathcal{O}$ and their policy is an implicit function of parameters θ_i:

$$\pi_i = \pi_{\theta_i} : \mathcal{S} \to \Delta(\mathcal{A}_i).$$

The agent does not have direct introspective access to the function π_{θ_i}. It can only *infer* a self-model $\hat{\pi}_i$ from its observations:

$$\hat{\pi}_i = f(o_1, a_1, r_1, \ldots, o_t) \approx \pi_i.$$

Definition 1.1 (Self-knowledge loss)**.** *The self-knowledge loss of agent i is:*

$$L_{self}^{(i)} = \mathbb{E}\big[\|\pi_i - \hat{\pi}_i\|^2\big]. \tag{1}$$

This measures the gap between "what I do" and "what I think I do."

Remark 1.2 (The "vibing" agent)**.** *An agent that performs well through habitual or intuitive action — what we call "vibing" — has a good policy π_i but a poor self-model $\hat{\pi}_i$. In flow-state terms (Parvizi-Wayne et al. 2024), the explicit self-model is attenuated while implicit performance is optimal. In Polanyi's terms, the agent has* tacit knowledge *that it cannot articulate.*

1.2 Evidence as Self-Knowledge

The self-knowledge loss is related to the evidence variance V_i from the EW-PG framework. Both are driven by the same information deficit: insufficient experience to resolve the $\mathcal{O} \to \Pi$ mapping.

Axiom 1.1 (Self-knowledge–evidence relationship)**.** *For each agent i with evidence variance $V_i > 0$:*

$$L_{self}^{(i)} \;\leq\; V_i. \tag{2}$$

Justification. By the data processing inequality, $I(\pi_i;\, \hat{\pi}_i) \leq I(\pi_i;\, \mathcal{O}_i)$. The evidence variance V_i is inversely related to $I(\pi_i;\, \mathcal{O}_i)$: high variance means the observation history is insufficient to pin down the value function, and *a fortiori* insufficient to pin down the policy. The bound (2) makes this precise.

2 The Communication Channel

2.1 End-to-End Communication

When agent i communicates its policy to teammate j, the information passes through three stages:

$$\underbrace{\pi_i}_{\text{true}} \xrightarrow{\text{self-model}} \underbrace{\hat{\pi}_i}_{\text{inferred}} \xrightarrow{\text{channel}} \underbrace{\tilde{\pi}_i^{(j)}}_{\text{received}} \tag{3}$$

Definition 2.1 (Communication loss)**.** *The total communication loss for agent j's model of agent i:*

$$L_{comm}^{(i\to j)} \;=\; \mathbb{E}\big[\|\pi_i - \tilde{\pi}_i^{(j)}\|^2\big]. \tag{4}$$

Theorem 2.2 (Loss decomposition)**.** *The communication loss decomposes as:*

$$L_{comm}^{(i\to j)} \;\leq\; 2\,L_{self}^{(i)} + 2\,L_{channel}^{(i\to j)} \tag{5}$$

where $L_{channel}^{(i\to j)} = \mathbb{E}[\|\hat{\pi}_i - \tilde{\pi}_i^{(j)}\|^2]$ is the pure channel distortion.

Proof. Write $\pi_i - \tilde{\pi}_i^{(j)} = (\pi_i - \hat{\pi}_i) + (\hat{\pi}_i - \tilde{\pi}_i^{(j)})$. By $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ (which follows from $(|a| - |b|)^2 \geq 0$), we get (5). □

Corollary 2.3 (Self-knowledge bottleneck)**.** *Even with a perfect channel ($L_{channel} = 0$):*

$$L_{comm}^{(i\to j)} \;\leq\; 2\,V_i. \tag{6}$$

An agent that doesn't know its own policy cannot communicate it.

2.2 Rate-Distortion with Self-Knowledge Floor

The communication problem connects to Shannon's rate-distortion theory. Agent i encodes $\hat{\pi}_i$ at rate R (bits per policy dimension). The channel's rate-distortion function $D_{\text{ch}}(R)$ gives the minimum achievable channel distortion at rate R.

The total distortion has a floor:

$$D_{\text{total}}(R) \;\geq\; \max\big(D_{\text{ch}}(R),\; L_{\text{self}}\big). \tag{7}$$

Remark 2.4 (Bandwidth vs. evidence)**.** *When $L_{self} > D_{ch}(R)$, the bottleneck is self-knowledge, not bandwidth. Increasing channel capacity does not help — the agent needs more* evidence*, not more bandwidth. The crossover rate $R^* : D_{ch}(R^*) = L_{self}$ defines sufficient bandwidth for the agent's evidence level.*

3 Coalition Formation

3.1 Coalition Rationality

Definition 3.1 (Coalition value)**.** *In an N-player stochastic game, the coalition value $V(S)$ for $S \subseteq \{1, \ldots, N\}$ is the joint expected payoff when members of S coordinate their policies.*

Definition 3.2 (Rational coalition)**.** *Coalition S is* rational *if:*

$$V(S) \;>\; \sum_{i \in S} V(\{i\}). \tag{8}$$

The excess $V(S) - \sum_i V(\{i\})$ is the coordination premium*.*

Remark 3.3 (Competitive coalitions)**.** *Condition* (8) *applies in* competitive *games too. In a 6-player game, a 3-player coalition may outperform three individuals playing against the opposing coalition. This is the formal version of "it pays to have allies even in competitive settings."*

3.2 Communication-Adjusted Coalition Value

The actual coalition value depends on communication quality:

$$V_{\text{comm}}(S) \;=\; V(S) - \underbrace{\sum_{i \in S} C_i \cdot L_{\text{comm}}^{(i)}}_{\text{coordination loss from imperfect communication}} \tag{9}$$

where C_i is the marginal coordination value of agent i.

Coalition S remains rational under imperfect communication iff:

$$V_{\text{comm}}(S) \;>\; \sum_{i \in S} V(\{i\}). \tag{10}$$

Using the loss decomposition (Theorem 2.2) and the evidence bound (Axiom 1.1):

$$V_{\text{comm}}(S) \;\geq\; V(S) - 2 \sum_{i \in S} C_i \, (V_i + L_{\text{ch}}^{(i)}). \tag{11}$$

High-evidence agents (low V_i) are more valuable coalition members.

4 The Cooperative Gradient

4.1 Update Rule

When agent i is in coalition S, its gradient is augmented with a cooperative correction based on communicated policies:

$$\pi_{i,n+1} = \text{proj}_{\Pi}\Big(\pi_{i,n} + \gamma_n \cdot w_i \cdot \big(\hat{v}_{i,n} + \beta_n \cdot \text{coop}_i(\pi_n, \tilde{\pi}_{S \setminus i})\big)\Big) \tag{12}$$

where:

- $w_i = V_{\min}/V_i$ is the evidence weight (from EW-PG),
- β_n is the cooperation strength schedule,

- coop_i is the cooperative correction from communicated policies.

Remark 4.1 (Triple-duty evidence weight)**.** *The weight w_i multiplies* both *the standard gradient and the cooperative term. This is correct: if agent i has poor self-knowledge (high V_i, low w_i), it should downweight its cooperative contribution too, since its self-model is unreliable. The evidence weight does triple duty:*

1. ***Gradient quality**: reduces PG variance (EW-PG, Theorem 5.1 of [**?**]).*
2. ***Self-knowledge**: bounds L_{self} (Axiom 1.1).*
3. ***Communication quality**: bounds L_{comm} (Corollary 2.3).*

All three are manifestations of the same underlying information content.

4.2 Communication Bias

The cooperative term uses $\tilde{\pi}_j$ (communicated) instead of π_j (true). This introduces bias:

$$\|\text{coop}(\pi, \pi_S) - \text{coop}(\pi, \tilde{\pi}_S)\| \;\leq\; L_{\text{coop}} \cdot \sqrt{L_{\text{comm}}} \;\leq\; L_{\text{coop}} \cdot \sqrt{2V_i + 2L_{\text{ch}}} \tag{13}$$

where L_{coop} is the Lipschitz constant of the cooperative term.

5 Convergence Analysis

5.1 Annealed Cooperation Preserves Convergence

Theorem 5.1 (Convergence of annealed cooperative PG)**.** *Under the conditions of Giannou et al. (2022), with:*

- $\gamma_n = \gamma/(n+m)^p$, $p \in (1/2, 1]$
- $\beta_n = \beta/(n+m)^r$, $r > 1 - p$
- *Cooperative term bounded:* $\|coop(\pi)\| \leq K$
- *Cooperative term vanishes at Nash:* $coop(\pi^*) = 0$

the cooperative PG (12) *converges to stable Nash at the same rate as standard PG.*

Proof sketch. Identical structure to annealed LOLA (OpponentShapingPG, Theorem 4.3). The cooperative term adds $\beta_n K$ to the bias bound: $\|b_n^{\text{coop}}\| \leq B_n + \beta_n K = O(n^{-\min(\ell_b, r)})$. The condition $r > 1 - p$ ensures $p + \min(\ell_b, r) > 1$, so Giannou's Theorem 1 applies with the augmented bias. □

5.2 Cooperative Basin Enlargement

Definition 5.2 (Cooperative reinforcement)**.** *The cooperative Hessian $H_C = \nabla_\pi\, coop(\pi^*)$ is* cooperatively reinforcing *if its symmetric part $S_C = (H_C + H_C^\top)/2$ is negative semi-definite.*

Theorem 5.3 (Basin enlargement)**.** *Under cooperative reinforcement, the cooperative PG has SOS parameter:*

$$\mu_{coop} = \mu + \beta \cdot \mu_C \tag{14}$$

where $-\mu_C$ is the largest eigenvalue of S_C (so $\mu_C \geq 0$). The basin of attraction is strictly larger when $\mu_C > 0$.

6 The Complete Ω-Gradient

Combining all three mechanisms, the full multi-agent gradient is:

$$\boxed{\pi_{i,n+1} = \mathrm{proj}_{\Pi}\Big(\pi_{i,n} + \gamma_n \cdot w_i \cdot \big(\underbrace{\hat{v}_{i,n}}_{\text{PG}} + \underbrace{\lambda_n \cdot \mathrm{OS}(\pi_n)}_{\text{LOLA}} + \underbrace{\beta_n \cdot \mathrm{coop}(\pi_n, \tilde{\pi}_S)}_{\text{coalition}}\big)\Big)} \tag{15}$$

Theorem 6.1 (Full Ω-PG convergence)**.** *The update* (15) *with annealed schedules* $\lambda_n, \beta_n \to 0$ *achieves:*

1. ***Variance improvement****:* $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ *from evidence weighting.*
2. ***Opponent-shaping basin****:* $\mu + \lambda\mu_H$ *from LOLA.*
3. ***Cooperative basin****:* $\mu + \beta\mu_C$ *from communication.*
4. ***Combined****:* $\mu_{full} = \mu + \lambda\mu_H + \beta\mu_C$, $C_{full} = \frac{\mathrm{HM}}{\mathrm{AM}} \cdot C_{std}$.

The three mechanisms are orthogonal: evidence weighting affects C*, opponent shaping and cooperation affect* μ *through independent Hessian contributions.*

6.1 Mapping to the Ω-Framework (theos eq. 27)

The five gradient components from the Ω-framework are now fully covered:

Component	Ω-framework	Formalization
(1) Exploration	$\mathbb{E}[L \cdot s_\theta]$	$\hat{v}_i$ (REINFORCE)
(2) Exploitation	$\mathbb{E}[\nabla_\theta L]$	$\hat{v}_i$ (backprop)
(3) Evidence	$-\mu V^{-2}\nabla V$	w_i (Keynesian weight)
(4) Alignment	∇D	$\beta_n \cdot \mathrm{coop}$
(5) Opp. shaping	$\sum_{j\neq i} \frac{\partial R_i}{\partial \theta_j}\frac{\partial \theta'_j}{\partial \theta_i}$	$\lambda_n \cdot \mathrm{OS}$

Component (4) was previously listed as "alignment/consensus" without formalization. The cooperative PG fills this gap: agents align their policies through lossy communication, with quality bounded by evidence weights.

7 The Six-Way Risk Decomposition — Complete

The Ω-framework decomposes multi-agent risk into six terms:

$$R_{\text{multi}} = \underbrace{R_{W\cap\Pi_N\cap B^c} + R_{S\cap\Pi_N\cap B^c}}_{\text{learnable}} + \underbrace{R_{W\cap\Pi_N\cap B} + R_{S\cap\Pi_N\cap B}}_{\text{Gödel-limited}} + \underbrace{R_{W\cap\Pi_U} + R_{S\cap\Pi_U}}_{\text{Keynes-limited}} \tag{16}$$

Method	Terms addressed	Mechanism
Standard PG	1–2 (learnable)	Gradient descent
EW-PG	+ 5–6 (Keynes)	Evidence-aware weighting
LOLA-PG	+ 3–4 (Gödel, adv.)	Infer opponent's blind spots
Coop-PG	+ 3–4 (Gödel, coop.)	Communicate own blind spots
Full Ω-PG	**All 6 terms**	**All mechanisms combined**

LOLA and cooperation address the Gödel-limited terms through *dual* mechanisms:

- **LOLA**: Agent i *infers* G_{F_j} from j's gradient response. (Competitive Gödelian step: I figure out what you can't see.)
- **Coop**: Agent j *communicates* (a lossy version of) G_{F_j}. (Cooperative Gödelian step: I tell you what I can see.)

These are independent information channels for the same Gödelian content. When both are available (intra-coalition + inter-coalition dynamics), they provide complementary information.

Only terms in $B_{\min}$ (the irreducible collective blind spot) remain permanently inaccessible — as Gödel guarantees.

8 The "Vibing" Spectrum and Evolutionary Pressure

The $\mathcal{O} \to \Pi$ mapping defines a spectrum:

$$\underbrace{\text{Full articulation}}_{\substack{L_{\text{self}}\approx 0 \\ \text{low } V_i \\ \text{Pearl Level 3}}} \quad \longleftrightarrow \quad \underbrace{\text{Pure vibing}}_{\substack{L_{\text{self}}\approx V_{\max} \\ \text{high } V_i \\ \text{Pearl Level 1}}}$$

The cooperative PG creates *evolutionary pressure toward articulability*: agents that can communicate their policies contribute more to coalitions, receive higher coalition payoff, and are preferred as coalition partners.

This is the formal version of the Ω-framework's claim that multi-agent interaction drives agents up the causal hierarchy: Pearl Level 1 $\to$ Level 2 $\to$ Level 3.

The pressure toward articulability is itself a *phase transition*: below a critical evidence threshold, agents cannot form effective coalitions and are stuck in competitive dynamics. Above the threshold, cooperation becomes possible, creating a positive feedback loop (more cooperation $\to$ more shared information $\to$ better evidence $\to$ better communication $\to$ more cooperation).

9 Implications for Competitive Environments

9.1 Dynamic Coalition Formation

In competitive games, coalitions are not fixed. At each stage, agents evaluate whether coalition membership improves their payoff:

$$i \in S_n^* \iff V_{\text{comm}}(S_n^*)/|S_n^*| > V(\{i\}) \tag{17}$$

where S_n^* is the optimal coalition at step n.

The communication quality (bounded by evidence) determines the *stable coalition structure*: agents with similar evidence levels form more effective coalitions (matched communication capacity).

9.2 The Competitive-Cooperative Duality

The full gradient (15) treats competition (LOLA) and cooperation (coop) symmetrically:

- Against opponents: LOLA term shapes their learning trajectory.
- With teammates: coop term aligns through communication.

- The evidence weight w_i moderates both.

An agent simultaneously cooperates within its coalition and competes against other coalitions. This is the generic structure of real-world multi-agent interaction: firms cooperate internally and compete externally; nations form alliances against common adversaries; individuals team up in games with shifting loyalties.

A Pseudocode

Algorithm 1: Cooperative PG with Lossy Communication (Coop-PG)

Input: Stochastic game $\mathcal{G}$, N agents
Input: Learning rates γ_n, cooperation schedule $\beta_n = \beta/(n+m)^r$
Input: Communication channel with capacity C, coalition update period T_{coal}
Output: Policies $\{\pi_i\}_{i=1}^N$ converging to stable Nash

Initialize $\pi_i \in \Pi$ for each agent i;
Initialize coalition structure $\mathcal{S} \leftarrow \{\{1\}, \{2\}, \ldots, \{N\}\}$;
Initialize self-models $\hat{\pi}_i \leftarrow \pi_i$ for each i;
for $n = 1, 2, \ldots$ **do**
  `// --- Phase 1: Standard gradient and evidence estimation ---`
  for $i = 1, \ldots, N$ **do**
    Collect trajectory, compute $\hat{v}_{i,n}$ and $\hat{V}_{i,n}$;
    $w_{i,n} \leftarrow V_{\min,n}/\hat{V}_{i,n}$;
  `// --- Phase 2: Self-model update (`$\mathcal{O} \to \Pi$`) ---`
  for $i = 1, \ldots, N$ **do**
    Update self-model from recent behavior:;
    $\hat{\pi}_{i,n} \leftarrow \textsc{InferPolicy}(o_{i,1:n}, a_{i,1:n})$;
    `// Self-knowledge loss:` $L^{(i)}_{\texttt{self}} \leq \hat{V}_{i,n}$
  `// --- Phase 3: Communication within coalitions ---`
  for *each coalition* $S \in \mathcal{S}$ **do**
    for $i \in S$ **do**
      Encode self-model: $s_i \leftarrow \textsc{Encode}(\hat{\pi}_{i,n}, C)$;
      Broadcast s_i to coalition members;
    for $i \in S$ **do**
      for $j \in S \setminus \{i\}$ **do**
        Decode teammate's signal:;
        $\tilde{\pi}_{j\to i} \leftarrow \textsc{Decode}(s_j)$;
        `// Total loss:` $\leq 2\hat{V}_{j,n} + 2L_{\texttt{ch}}$
  `// --- Phase 4: Cooperative gradient ---`
  for $i = 1, \ldots, N$ **do**
    Let S_i be the coalition containing i;
    if $|S_i| > 1$ **then**
      Compute cooperative correction from communicated policies:;
      $\text{coop}_{i,n} \leftarrow \nabla_{\theta_i} \hat{R}_{S_i}(\pi_{i,n}, \tilde{\pi}_{S_i \setminus i})$;
    else
      $\text{coop}_{i,n} \leftarrow 0$;
    $\pi_{i,n+1} \leftarrow \text{proj}_\Pi\big(\pi_{i,n} + \gamma_n \cdot w_{i,n} \cdot (\hat{v}_{i,n} + \beta_n \cdot \text{coop}_{i,n})\big)$;
  `// --- Phase 5: Coalition update (every` $T_{\texttt{coal}}$ `steps) ---`
  if $n \bmod T_{coal} = 0$ **then**
    for *each pair* i, j *in different coalitions* **do**
      Estimate coalition value $\hat{V}(S_i \cup S_j)$ from recent data;
      if $\hat{V}(S_i \cup S_j) > \hat{V}(S_i) + \hat{V}(S_j)$ **then**
        Merge: $\mathcal{S} \leftarrow \mathcal{S} \setminus \{S_i, S_j\} \cup \{S_i \cup S_j\}$;
    for *each coalition* S *with* $|S| > 1$ **do**

Algorithm 2: Full Ω-PG (EW-LOLA-Coop)

Input: All inputs from EW-PG, LOLA-PG, Coop-PG
Input: Shaping schedule λ_n, cooperation schedule β_n
Output: Policies converging to stable Nash with all three improvements

Initialize policies, variances, opponent models, self-models, coalitions;
for $n = 1, 2, \ldots$ **do**
  `// Compute all components`
  $\hat{v}_{i,n},\ w_{i,n}$ for each i ; `// EW-PG (gradient + evidence)`
  $\mathrm{OS}_{i,n}$ for each i ; `// LOLA (opponent shaping)`
  $\tilde{\pi}_{j\to i}$ for $j \in S_i$; `// Communication`
  $\mathrm{coop}_{i,n}$ for each i ; `// Cooperative correction`
  `// Full Ω-update (theos eq. 27, all 5 components)`
  for $i = 1, \ldots, N$ **do**
    $g_{i,n} \leftarrow \hat{v}_{i,n} + \lambda_n \cdot \mathrm{OS}_{i,n} + \beta_n \cdot \mathrm{coop}_{i,n}$;
    $\pi_{i,n+1} \leftarrow \mathrm{proj}_{\Pi}\big(\pi_{i,n} + \gamma_n \cdot w_{i,n} \cdot g_{i,n}\big)$;
  Update coalitions every T_{coal} steps ; `// Phase 5 of Alg. 1`
`// Convergence guarantees (Theorem 6.1):`
`//` $[\|\pi_n - \pi^*\|^2] = O\big(\frac{\mathrm{HM}(V)}{\mathrm{AM}(V)} \cdot C_{\mathtt{std}} \,/\, n^q\big)$
`//` $\mu_{\mathtt{full}} = \mu + \lambda\mu_H + \beta\mu_C$

Remark A.1 (Implementation notes). *1.* InferPolicy *reconstructs $\hat{\pi}_i$ from observation-action history. Options: (a) frequency counts (tabular), (b) behavioral cloning network trained on own (s, a) pairs, (c) inverse RL.*

2. Encode/Decode *implement lossy compression of the self-model. For continuous policies: quantize and transmit parameters. For tabular: transmit action distributions with precision $\propto C$.*

3. Coalition updates (Phase 5) are expensive. The period T_{coal} should be large enough for value estimates to stabilize (e.g., 100–1000 steps).

4. In competitive games, coalitions are temporary. An agent defects when its individual value exceeds its per-capita coalition share. The split check (Phase 5) handles this.

5. The self-knowledge loss $L^{(i)}_{self} \leq \hat{V}_{i,n}$ is monitored but not explicitly minimized. It decreases naturally as the agent gathers more evidence. To accelerate: add an intrinsic reward for self-model accuracy (curiosity about own policy).